Adaptive Finite Time Output Feedback Bipartite Tracking Control for Nonlinear Multiagent Systems

Xinjun Wang, Ben Niu, Yahui Gao, and Zihao Shang

***Abstract*— This paper investigates event-triggered based finite-time bipartite consensus tracking control problem for multi-agent systems (MASs) over signed directed graphs. Compared with the existing related results, the main features of the results presented in this paper are as follows: (*i*) The strict limitation for nonlinear MASs with a structurally balanced digraph is removed by introducing a prioritized strategy. (*ii*) A state estimator is constructed by combining neural networks (NNs), which reconstructs the immeasurable system states of each agent in nonstrict-feedback form for the first time and approximates the completely unknown nonlinearities that exist in the system. (*iii*) A new distributed control algorithm, named finite-time distributed control strategy is designed by introducing a novel first-order filter, which ensures that the consensus errors can obtain a fast convergence. (*iv*) The results are successfully extended to the event-based bipartite tracking control for MASs by using relative threshold strategy. Moreover, the proposed control algorithm is applied to the consensus problem of a group of forced damped pendulums (FDPs). Based on backstepping method and Lyapunov stability theory, the distributed consensus tracking errors can converge to a small neighborhood of the origin at a finite-time under the proposed protocol, and all the signals of the closed-loop system remain bounded. Finally, the simulation results illustrate the validity of the proposed schemes.**

***Note to Practitioners*—The studies of the bipartite consensus problems of multi-agent systems (MASs) have gained great attention due to its applications in multi-spacecraft systems, natural networks, and biological communities, and a new event-triggered based finite time bipartite tracking consensus control method for multi-agent systems (MASs) is studied in this paper. In practical applications, the finite time method can obtain the optimal control performance in time. Then, in order to save the communication and computation resources for the nonlinear MASs, this paper extends the proposed control strategy to the event-based bipartite tracking control problem for MASs by using the relative threshold event-triggered rules, which greatly conserves the communication resources between the control signal and the actuator. In addition, the dynamic model and backstepping technology used in this paper are general and practical.**

***Index Terms*— Multi-agent systems, prioritized strategy, bipartite consensus, adaptive neural control, state observer.**

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I. Introduction

In RECENT years, the studies of the cooperative consensus problems of multi-agent systems (MASs) have gained great attention due to its applications in multi-spacecraft systems, natural networks, and biological communities, for instance, [1], [2], [3] and the references therein. Given their widespread applications, consensus problem has been widely studied and a variety of results over nonnegative graphs have been explored in [4], [5], and [6]. A nonnegative graph means that the weights of the edges of graph are nonnegative and the interactions among agents are assumed to completely cooperation. However, antagonistic interactions indeed exist in numerous situations. For example, some potential applications are competition robotic soccer [7], multisensor networks [8], opinion dynamics [9], where the communications are illustrated via signed graphs with positive (negative) adjacency matrix, simultaneously.

Recently, researchers are no longer limited to the cooperative MASs, and the control strategies of bipartite consensus tracking of MASs over signed graphs have been investigated in [10], [11], [12], and [13]. The attribute "bipartite" means that their interactions can be denoted by a signed graph, whose edges' weights may be negative or positive. For example, a bipartite consensus protocol with Laplacian feedback design methods for MASs was proposed in [8]. Moreover, the bipartite consensus control method for generic linear MASs over signed digraphs is studied in [11], [12], and [14]. It should be pointed out that the above results are only considered the linear integrator dynamics. However, in many actual applications, MASs are inherently nonlinear and uncertain dynamics [15], [16], [17]. Besides, great attention and developments have been gained in neural networks (NNs) or fuzzy systems based control design for nonlinear systems [18], [19], [20] where NNs are used to deal with unknown nonlinearities. Thus, the above results were extended to second-order nonlinear MASs in [21]. Nevertheless, the proposed method in [21] is based on the condition that the signed graph must be structurally balanced, and bipartite consensus of MASs without using the structurally balanced condition of graph has not been achieved.

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On the other hand, the aforementioned works are mainly concerned with the infinite-time stability, i.e., the bipartite tracking errors converge to a small region only when time goes to infinity, which cannot satisfy fast convergence rate of systems [22]. Instead, the finite-time control method [23], [24], [25] can rapidly make system transient performances achieved and has the shorter convergent time. Authors in [26] and [27] studied finite-time bipartite consensus issues for MASs. However, the works in [26] and [27] only focus on the second-order MASs, while a few works have been investigated for high-order systems [28]. Authors in [29] adopted finite-time consensus problem for nonlinear MASs with matching conditions. Recently, the problem of adaptive finite-time bipartite consensus tracking was studied for nonlinear MASs in [30] and [31]. However, the above results require that the signed graph must be structurally balanced. It is not a trivial matter to achieve finite-time bipartite consensus without using the structurally balanced condition of graph for uncertain nonlinear MASs.

In addition, with the rapid development of networked systems, the emergence of event-triggered control has provided an appealing avenue for effective transmissions of measured signals. Early results of event-triggered control mainly focus on linear systems, see [32], [33], and [34] and the references therein. Meanwhile, some significant results based on event-triggered have been obtained for different types of nonlinear systems [35], [36], [37], [38], [39], [40], [41]. In addition, a distributed event-triggered consensus control protocol was studied for general linear MASs under directed graph in [42], and the result was extended to nonlinear MASs in [43]. It is notable that the above results [37], [38], [39], [40], [41] were assumed that all the systems states are measurable. However, it is difficult to measure all the states of the systems in many actual applications. Thus, it is more interesting to focus on the event-based output feedback control problem of nonlinear systems [44]. Recently, the authors in [45] investigated the distributed observer based cooperative output regulation problem for linear MASs. The authors in [46] proposed the adaptive fuzzy output consensus tracking control approach for nonlinear MASs. To the best of our knowledge, the event-triggered bipartite tracking control problem for nonlinear MASs with the signed graph to be structurally unbalanced is yet to be investigated.

The above discussion motivates us for this study. This paper is concerned with the neural state observer based event-triggered finite-time bipartite tracking control problem for nonlinear MASs. By proposing a prioritized strategy, a directed signed graph only containing a spanning tree is taken into account. On this basis, a distributed neural-based dynamic surface control law is proposed to guarantee all the signals in the closed-loop system are bounded. The main contributions of our work which are showed as follows.

(1) A more general condition that a signed digraph only containing a spanning tree is considered in this paper, while the strict limitation to a signed digraph be structurally balanced is removed. However, such a relaxation on the condition of a graph also faces a major issue: How to classify all agents into two different sets such that the agents can successfully achieve a bipartition? In order to solve the question, based on the idea of a shortest path, a prioritized strategy is introduced to achieve a bipartition of the agents in this paper. On this basis, a bipartite tracking control protocol is proposed for the nonlinear MASs and the bipartite consensus is successfully achieved.

(2) By using the relative threshold event-triggered rules and backstepping technology, this paper is extended to the event-based bipartite tracking control protocol for MASs, which greatly conserves the communication resources between the control signal and the actuator.

(3) Only the output information is available in designing the control protocol in this paper, and the neural networks based state estimator is developed to estimate the unmeasured states. Then, a new neural observer based adaptive bipartite control protocol is given, in which the uncertain dynamics are approximated by using the NNs. Besides, the distributed consensus tracking errors can converge to a small neighborhood of the origin at a finite-time under the proposed protocol and the "explosion of complexity" is alleviated based dynamic surface method in this paper.

II. System Descriptions

A. Graph Theory

Let $G \triangleq \{O, E, A\}$ denote a weighted signed directed graph (diagraph for short) of order N, $O = \{o_i : 1, \ldots, N\}$ represents a set of follower agents, $E \subseteq \{e_{ij} : i = 1, \ldots, N, j = 1, \ldots, N, i \neq j\}$ denotes a set of edges, and adjacency matrix is described as $A = [a_{ij}] \in R^{N\times N}$, where $a_{ij} < 0$ means the antagonistic interaction and $a_{ij} > 0$ the cooperative interaction; otherwise $a_{ij} = 0$. The set of neighbors of ith follower is $N_i = \{j \mid a_{ij} \neq 0\}$. Laplacian matrix is defined as $L = D - A$, where $L \in R^{N\times N}$, $D = \text{diag}(d_i)$ is the in-degree matrix of ith follower, and $d_i = \sum_{j\in N_i} |a_{ij}|$.

Assumption 1 [29]: The signed digraph G has a directed spanning tree with the leader as its root node.

Assumption 2: The output of leader $r(t)$ is a smooth bounded function, there exists a positive constant Y_0 such that $|r(t)| \leq Y_0$.

Definition 1: If the agents $o_1, o_p \in E$, a shortest path $P_1 : o_p \to o_1$ (o_1 can obtain the information from o_p) is defined:

$$P_1 = \min\{(o_1, o_2), (o_2, o_3), \ldots, (o_{p-1}, o_p)\} \subset E \quad (1)$$

A path $P_1 > 0$ if the edge weights $a_{1,2} \cdot a_{2,3} \cdots a_{p-1,p} > 0$; $P_1 < 0$ if $a_{1,2} \cdot a_{2,3} \cdots a_{p-1,p} < 0$. S_{P_1} is defined

$$S_{P_1} = \begin{cases} 1, & P_1 > 0 \\ -1 & P_1 < 0 \end{cases} \quad (2)$$

and $L_{P_1} = p - 1$ denotes length of P_1.

B. Prioritized Strategy

In order to achieve a bipartition for a network satisfying Assumption 1, a prioritized strategy [10] is introduced according to the concept of shortest path P. For any node o_i, we can find at least one shortest path $P_{i,1} : o_0 \to o_i$ (o_i can obtain the information from leader o_0). On the other hand, we define a set $\Omega_0 = \{o_0\}$, $\Omega_1 = \{o_i | L_{P_i} = 1\}$, $\Omega_2 = \{o_i | L_{P_i} = 2\}$, …, $\Omega_l = \{o_i | L_{P_i} = l\}$.

In a shortest path, if $o_i \in \Omega_k$, $o_j \in \Omega_{k-1}$, $(1 \leq k \leq l)$, then o_j is called a parent node for o_i, and o_i is called a child node

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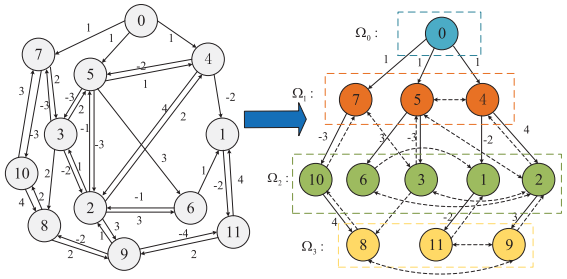


Fig. 1. Bipartition process by using prioritized strategy.

for o_j. An unique parent node o_j for node o_i can be chosen as $j = \min\{\max\{|a_{i,j}|, o_i \in \Omega_k, o_j \in \Omega_{k-1}\}\}$, if $\max\{|a_{i,j_1}| = |a_{i,j_2}|, o_i \in \Omega_k, o_{j_1}, o_{j_2} \in \Omega_{k-1}\}$, then an unique parent node o_j for node o_i can be chosen as $j = \min\{j_1, j_2\}$.

For example, by using prioritized strategy, a shortest path for each agent can be uniquely determined, and all the nodes can be achieved a bipartition of the sets O_1 and O_2. In this paper, we define $o_0 \in O_1$. For agent $o_i, i = 1, \ldots, 11$ (see Fig.1), if $S_{P_i} = 1$, then $o_i \in O_1$, else, $o_i \in O_2$. Clearly, $o_0, o_2, o_4, o_5, o_6, o_7, o_9, o_{11} \in O_1$, and $o_1, o_3, o_8, o_{10} \in O_2$. Then, a bipartition is successfully achieved in this paper.

C. Model Description

Consider the ith follower agent with the following dynamics:

$$\begin{aligned} \dot{x}_{i,k} &= x_{i,k+1} + f_{i,k}(\bar{x}_{i,k}) \\ \dot{x}_{i,n_i} &= u_i + f_{i,n_i}(\bar{x}_i) \\ y_i &= x_{i,1} \end{aligned} \tag{3}$$

where $k = 1, \ldots, n_i - 1$; $\bar{x}_i = [x_{i,1}, x_{i,2}, \ldots, x_{i,n_i}]^T \in R^{n_i}$, $(i = 1, \ldots, N)$ stands for the system state of the ith follower, $y_i \in R$ is the system output. $f_{i,m}(\cdot)$ are unknown smooth functions. $u_i \in R$ is the control protocol.

Lemma 1 [47]: For any real numbers $\kappa_1 > 0$, $\kappa_2 > 0$, $0 < \eta < 1$, an extended Lyapunov condition of finite-time stability can be given $\dot{V}(x) + \kappa_1 V(x) + \kappa_2 V^{\eta}(x) \le 0$, where $T_r \le t_0 + [1/(\kappa_1(1-\eta))]\ln[(\kappa_1 V^{1-\eta}(t_0) + \kappa_2)/\kappa_2]$.

Assumption 3: There exists a set of constant $B_{i,q}$, $q = 1, \ldots, n_i$ for $\forall X_1, X_2 \in R^i$ such that the following inequality holds:

$$\left|f_{i,q}(X_1) - f_{i,q}(X_2)\right| \le B_{i,q}\left\|X_1 - X_2\right\| \tag{4}$$

where $\|\cdot\|$ represents the 2-norm.

D. Neural Networks

Consider the Gaussian basis function [48]:

$$s_j(Z) = \exp(-(Z - C_j)^T(Z - C_j)/2b_j^2) \tag{5}$$

where b_j is the width of the Gaussian function, and $C_j = [C_{j1}, C_{j2}, \ldots, C_{jq}]^T$ is the center vector.

A RBF NNs can estimate a continuous function $f(Z)$ on a compact set $Z \in \Omega_Z \subset R^q$ with the neural node number $l > 1$ [49]:

$$f(Z) = \Phi^T S(Z) + \delta(Z) \tag{6}$$

where $S(Z) = [s_1(Z), s_2(Z), \ldots, s_l(Z)]^T$ is the basis function vector, $\Phi = [\phi_1, \phi_2, \ldots, \phi_l]^T$ is the ideal constant weight vector, $\delta(Z)$ denotes approximation error satisfies $|\delta(Z)| \le \xi$ with $\xi > 0$.

III. Main Results

In this section, we will construct a state estimator for multiagent systems (3). Notice that the state variables $x_{i,k}$ $(i = 1, \ldots, N, k = 2, \ldots, n_i)$ are possibly unavailable. Firstly, the system (3) can be rewritten as:

$$\begin{aligned} \dot{x}_i &= A_i x_i + \sum_{q=1}^{n_i} K_{i,q}(f_{i,q}(\hat{\bar{x}}_{i,q}) + \Delta f_{i,q}) + D_i u_i \\ y_i &= C_i^T x_i \end{aligned} \tag{7}$$

where $\Delta f_{i,q} = f_i(\bar{x}_{i,q}) - f_i(\hat{\bar{x}}_{i,q})$, $\hat{\bar{x}}_i = [\hat{x}_{i,1}, \ldots, \hat{x}_{i,n_i}]^T$ is the estimate of x_i,

$$A_i = \begin{bmatrix} 0 & 1 & & & \\ 0 & 0 & 1 & & \\ \vdots & \vdots & & \ddots & \\ 0 & 0 & \ldots & 0 & 1 \\ 0 & 0 & \cdots & 0 & 0 \end{bmatrix},$$

$$K_{i,1} = [1, 0, \ldots, 0]^T, \ldots, K_{i,n_i} = [0, 0, \ldots, 1]^T,$$

$$D_i = \left[0, 0, \ldots, 1\right]^T, \quad C_i^T = \left[1, 0, \ldots, 0\right].$$

The NNs is introduced to approximate the function $f_{i,q}(\bar{x}_{i,q})$, i.e., $f_{i,q}(\bar{x}_{i,q}) = \Phi_{i,q}^T S_{i,q}(\bar{x}_{i,q}) + \delta_{i,q}$, where $\delta_{i,q}$ is the approximation error, $|\delta_{i,q}| \le \xi_{i,q}$. Thus, a state estimator is designed as follows:

$$\begin{aligned} \dot{\hat{x}}_i &= A_{i0}\hat{x}_i + \sum_{q=1}^{n_i} K_{i,q}\hat{f}_{i,q}(\hat{\bar{x}}_{i,q}) + D_i u_i + L_i y_i \\ \hat{y}_i &= C_i^T \hat{x}_i \end{aligned} \tag{8}$$

where $A_{i0} = A_i - L_i C_i^T$, $L_i = \left[l_{i,1}, l_{i,2}, \ldots, l_{i,n_i}\right]^T$, $\hat{f}_{i,q}(\hat{\bar{x}}_{i,q}) = \hat{\Phi}_{i,q}^T S_{i,q}(\hat{\bar{x}}_{i,q})$. L_i are the design parameters such that the matrix A_{i0} is a strict Hurwitz matrix, which means for a given matrix $Q_i > 0$, there exists a matrix $P_i > 0$ satisfying the following equation:

$$A_{i0}^T P_i + P_i A_{i0} = -Q_i \tag{9}$$

Define the estimation errors as $\bar{e}_{i,q} = x_{i,q} - \hat{x}_{i,q}$. Then we can obtain the observer error equation as:

$$\dot{\bar{e}}_i = A_{i0}\bar{e}_i + \sum_{q=1}^{n_i} K_{i,q}(\tilde{\Phi}_{i,q}^T S_{i,q}(\hat{\bar{x}}_{i,q}) + \delta_{i,q} + \Delta f_{i,q}) \tag{10}$$

To consider the performance of the state estimator (8), the following Theorem is proposed.

Theorem 1: [Boundedness of Local State Observer] For the MASs includes of N follower agents (3) and a leader $r(t)$, if $\lambda_{min}(Q_i) - 3n_i - \frac{5}{2} - \|P_i\|^2 \sum_{q=1}^{n_i} B_{i,q}^2 > 0$ and Lemma 2 holds, the proposed state observer (8) guarantees the estimation errors of follower agents converge to a specified neighborhood around the origin as

$$\left\|\bar{e}_i\right\|^2 \le \frac{b_0}{\lambda_{min}(P_i)a_0} \tag{11}$$

Proof: The Lyapunov candidate for the observer error surfaces is designed as below

$$V_0 = \sum_{i=1}^{N} \bar{e}_i^T P_i \bar{e}_i \tag{12}$$

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The derivative of V_0 is given as

$$\dot{V}_0 = -\bar{e}_i^T Q_i \bar{e}_i + 2\bar{e}_i^T P_i \left(\sum_{q=1}^{n_i} K_{i,q} (\tilde{\Phi}_{i,q}^T S_{i,q}(\hat{\bar{x}}_{i,q}) + \delta_{i,q} + \Delta f_{i,q}) \right) \tag{13}$$

Using Young's inequality, one has

$$2\bar{e}_i^T P_i \sum_{q=1}^{n_i} K_{i,q} (\tilde{\Phi}_{i,q}^T S_{i,q}(\hat{\bar{x}}_{i,q}) + \delta_{i,q}) \leq 2(n_i+1)\|\bar{e}_i\|^2 + \|P_i\|^2 \sum_{q=1}^{n_i} (\tilde{\Phi}_{i,q}^2 + \xi_{i,q}^2) \tag{14}$$

$$2\bar{e}_i^T P_i \sum_{q=1}^{n_i} K_{i,q} \Delta f_{i,q} \leq \left(n_i + \|P_i\|^2 \sum_{q=1}^{n_i} B_{i,q}^2 \right) \|\bar{e}_i\|^2 \tag{15}$$

Substituting (14) - (15) into (13) results in

$$\begin{aligned} \dot{V}_0 &\leq -\left(\lambda_{min}(Q_i) - 3n_i - \frac{5}{2} - \|P_i\|^2 \sum_{q=1}^{n_i} B_{i,q}^2 \right) \|\bar{e}_i\|^2 \\ &\quad + \|P_i\|^2 \sum_{q=1}^{n_i} (\tilde{\Phi}_{i,q}^2 + \xi_{i,q}^2) \\ &\leq -a_0 V_0 + b_0 \end{aligned} \tag{16}$$

where $a_0 = \min_{i=1,\ldots,N} \left\{ \frac{\lambda_{min}(Q_i) - 3n_i - \frac{5}{2} - \|P_i\|^2 \sum_{q=1}^{n_i} B_{i,q}^2}{\lambda_{max}(P_i)} \right\}$ and $b_0 = \sum_{i=1}^{N} \left\{ \|P_i\|^2 \sum_{q=1}^{n_i} (\tilde{\Phi}_{i,q}^2 + \xi_{i,q}^2) \right\}$. Hence, (11) can be acquired. ■

Remark 1: In this paper, Q_i, P_i, $B_{i,q}$ are the design parameters, which should be chosen such that $\lambda_{min}(Q_i) - 3n_i - \frac{5}{2} - \|P_i\|^2 \sum_{q=1}^{n_i} B_{i,q}^2 > 0$.

Consider node o_i ($i \in \Omega_k, k \geq 1$) with its parent node o_j ($j = \min\{\max\{|a_{i,j}|, o_j \in \Omega_{k-1}\}\}$) in path P_i, the following error variables are defined as

$$\begin{aligned} e_{i,1} &= x_{i,1} - \text{sign}(a_{ij}) x_{j,1} \\ e_{i,k} &= \hat{x}_{i,k} - \pi_{i,k}, \quad k = 2, \ldots, n_i \\ \chi_{i,k} &= \pi_{i,k} - \alpha_{i,k-1}, \quad k = 2, \ldots, n_i \end{aligned} \tag{17}$$

where $\pi_{i,k}$ is the output of a first-order filter, $\alpha_{i,k-1}$ is virtual control protocol, $\chi_{i,k}$ is boundary error surface. A first-order filter is defined as

$$\varpi_{i,k}\dot{\pi}_{i,k} + \pi_{i,k} = \alpha_{i,k-1} + \text{sig}^{\gamma}(-\chi_{i,k}), \quad \pi_{i,k}(0) = \alpha_{i,k-1}(0) \quad \text{for } i = 1, 2, \ldots, N \text{ and } k = 2, \ldots, n_i \tag{18}$$

where $\text{sig}^{\gamma}(x) = \text{sign}(x)|x|^{\gamma}$, $\gamma \geq 0$, $x \in R$, $\varpi_{i,k} > 0$ is the time constant.

The protocol is first proposed as

$$\begin{aligned} \alpha_{i,1} &= (-k_{i,1} e_{i,1} - 5e_{i,1} - r_{i,1}\text{sig}^{\gamma}(e_{i,1}) + \hat{\Delta}_{i,1}^T \bar{S}_{i,1}) \\ &\quad - \hat{\Phi}_{i,1}^T S_{i,1} \end{aligned} \tag{19}$$

$$\begin{aligned} \alpha_{i,q} &= -k_{i,q} e_{i,q} - \frac{5}{2} e_{i,q} - r_{i,q}\text{sig}^{\gamma}(e_{i,q}) - \hat{\Phi}_{i,q}^T S_{i,q} \\ &\quad - l_{i,q}\bar{e}_{i,1} + \dot{\pi}_{i,q} \quad \text{for} \quad q = 2, \ldots, n_i - 1 \end{aligned} \tag{20}$$

$$\begin{aligned} \alpha_{i,n_i} &= -k_{i,n_i} e_{i,n_i} - \frac{3}{2} e_{i,n_i} - r_{i,n_i}\text{sig}^{\gamma}(e_{i,n_i}) - \hat{\Phi}_{i,n_i}^T S_{i,n_i} \\ &\quad - l_{i,n_i}\bar{e}_{i,1} + \dot{\pi}_{i,n_i} \end{aligned} \tag{21}$$

where $k_{i,q}$, $r_{i,q}$ are positive design parameters, and $0 < \gamma < 1$, $\hat{\Delta}_{i,1}$, $\hat{\Phi}_{i,q}$ are the estimated values of NNs weight vector, for $i = 1, \ldots, N$ and $q = 1, \ldots, n_i$.

Theorem 2: (Output-feedback bipartite consensus stability) For a multi-agent network (3) containing a spanning tree. Let Assumption 1 hold, and suppose the event-triggered mechansim (41) and the controller (45), adaptive laws (28), (29), (36), (50), the virtual controller (19), (20) and (21) are designed. Then, all the signals of the closed-loop systems are bounded and the bipartite tracking error can converge to a small neighborhood of origin in finite-time.

Proof: To derive the main results, we perform the recursive control design as following steps.

Step 1:

$$V_{i,1} = \frac{1}{2} e_{i,1}^2 + \frac{1}{2}\tilde{\Phi}_{i,1}^2 + \frac{1}{2}\tilde{\Delta}_{i,1}^2 \tag{22}$$

where $\tilde{\Phi}_{i,1} = \Phi_{i,1} - \hat{\Phi}_{i,1}$, $\tilde{\Delta}_{i,1} = \Delta_{i,1} - \hat{\Delta}_{i,1}$ are the estimated errors of NNs weight vector.

The derivative of $V_{i,1}$ is given as

$$\begin{aligned} \dot{V}_{i,1} &= e_{i,1}\big(\bar{e}_{i,2} + e_{i,2} + \chi_{i,2} + \alpha_{i,1} + f_{i,1}(x_{i,1}) \\ &\quad - \text{sig}(a_{ij})(\hat{x}_{j,2} + \bar{e}_{j,2} + f_{j,1}(x_{j,1}))\big) \\ &\quad - \tilde{\Phi}_{i,1}\dot{\hat{\Phi}}_{i,1} - \tilde{\Delta}_{i,1}\dot{\hat{\Delta}}_{i,1} \end{aligned} \tag{23}$$

where $\bar{f}_{j,1} = -\text{sig}(a_{ij})(f_{j,1}(\bar{x}_{j,1}) + \bar{e}_{j,2})$. A NNs is introduced to approximate the function $\bar{f}_{j,1}$, i.e., $\bar{f}_{j,1} = \Delta_{i,1}^T \bar{S}_{i,1} + \bar{\delta}_{i,1}$, where $\bar{\delta}_{i,1}$ is the approximation error, $|\bar{\delta}_{i,1}| \leq \bar{\xi}_{i,1}$.

Using Young's inequality, one has

$$e_{i,1}(\bar{e}_{i,2} + e_{i,2} + \chi_{i,2}) \leq \frac{3}{2} e_{i,1}^2 + \frac{1}{2} e_{i,2}^2 + \frac{1}{2}\|\bar{e}_i\|^2 + \frac{1}{2}\chi_{i,2}^2 \tag{24}$$

$$e_{i,1}\delta_{i,1} \leq \frac{1}{2} e_{i,1}^2 + \frac{1}{2}\xi_{i,1}^2 \tag{25}$$

$$-e_{i,1}\bar{\delta}_{i,1} \leq \frac{1}{2} e_{i,1}^2 + \frac{1}{2}\bar{\xi}_{i,1}^2 \tag{26}$$

Substituting (19), (24) - (26) into (23) results in

$$\begin{aligned} \dot{V}_{i,1} &\leq -k_{i,1} e_{i,1}^2 - e_{i,1} r_{i,1}\text{sig}^{\gamma}(e_{i,1}) + e_{i,1}\big(\tilde{\Phi}_{i,1}^T S_{i,1} \\ &\quad - \tilde{\Delta}_{i,1}^T \bar{S}_{i,1}\big) + \frac{1}{2} e_{i,2}^2 + \frac{1}{2}\chi_{i,2}^2 + \frac{1}{2}\xi_{i,1}^2 \\ &\quad + \frac{1}{2}\bar{\xi}_{i,1}^2 - \tilde{\Phi}_{i,1}\dot{\hat{\Phi}}_{i,1} - \tilde{\Delta}_{i,1}\dot{\hat{\Delta}}_{i,1} \end{aligned} \tag{27}$$

and the adaptive laws of $\dot{\hat{\Phi}}_{i,1}$ and $\dot{\hat{\Delta}}_{i,1}$ are designed as

$$\dot{\hat{\Phi}}_{i,1} = e_{i,1} S_{i,1} - \Gamma_{i,1}\hat{\Phi}_{i,1} \tag{28}$$

$$\dot{\hat{\Delta}}_{i,1} = -(e_{i,1}\bar{S}_{i,1} + \bar{\Gamma}_{i,1}\hat{\Delta}_{i,1}) \tag{29}$$

where $\Gamma_{i,1}$, $\bar{\Gamma}_{i,1}$ are positive design constants. Substituting (28) and (29) into (27), one has

$$\begin{aligned} \dot{V}_{i,1} &\leq -k_{i,1} e_{i,1}^2 - e_{i,1} r_{i,1}\text{sig}^{\gamma}(e_{i,1}) + e_{i,1}(\tilde{\Phi}_{i,1}^T S_{i,1} \\ &\quad - \tilde{\Delta}_{i,1}^T \bar{S}_{i,1}) + \frac{1}{2} e_{i,2}^2 + \frac{1}{2}\chi_{i,2}^2 + \frac{1}{2}\xi_{i,1}^2 \\ &\quad + \frac{1}{2}\bar{\xi}_{i,1}^2 + \Gamma_{i,1}\tilde{\Phi}_{i,1}^T\hat{\Phi}_{i,1} + \bar{\Gamma}_{i,1}\tilde{\Delta}_{i,1}^T\hat{\Delta}_{i,1} \end{aligned} \tag{30}$$

Step k ($k = 2, \ldots, n_i - 1$) Define

$$V_{i,k} = \frac{1}{2} e_{i,k}^2 + \frac{1}{2}\chi_{i,k}^2 + \frac{1}{2}\tilde{\Phi}_{i,k}^2 \tag{31}$$

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where $\tilde{\Phi}_{i,k} = \Phi_{i,k} - \hat{\Phi}_{i,k}$ are the estimation errors of NNs weight vector.

$$\begin{aligned}\dot{V}_{i,k} &= e_{i,k}\dot{e}_{i,k} + \chi_{i,k}\dot{\chi}_{i,k} - \tilde{\Phi}_{i,k}\dot{\hat{\Phi}}_{i,k}\\ &= e_{i,k}(\hat{x}_{i,k+1} + \hat{\Phi}_{i,k}^T S_{i,k} + \tilde{\Phi}_{i,k}S_{i,k} - \tilde{\Phi}_{i,k}S_{i,k} + l_{i,k}\bar{e}_{i,1}\\ &\quad - \dot{\pi}_{i,k}) + \chi_{i,k}\dot{\chi}_{i,k} - \tilde{\Phi}_{i,k}\dot{\hat{\Phi}}_{i,k}\\ &= e_{i,k}(e_{i,k+1} + \chi_{i,k+1} + \alpha_{i,k} + \hat{\Phi}_{i,k}^T S_{i,k} + \tilde{\Phi}_{i,k}S_{i,k}\\ &\quad - \tilde{\Phi}_{i,k}S_{i,k} + l_{i,k}\bar{e}_{i,1} - \dot{\pi}_{i,k}) + \chi_{i,k}\dot{\chi}_{i,k}\\ &\quad - \tilde{\Phi}_{i,k}\dot{\hat{\Phi}}_{i,k}\end{aligned} \tag{32}$$

Consider

$$-e_{i,k}\tilde{\Phi}_{i,k}S_{i,k} \le e_{i,k}^2 + \frac{1}{4}\tilde{\Phi}_{i,k}^2 \tag{33}$$

$$e_{i,k}(e_{i,k+1} + \chi_{i,k+1}) \le e_{i,k}^2 + \frac{1}{2}e_{i,k+1}^2 + \frac{1}{2}\chi_{i,k+1}^2 \tag{34}$$

Substituting (20) into (32) results in

$$\begin{aligned}\dot{V}_{i,k} &= e_{i,k}(-k_{i,k}e_{i,k} - \frac{1}{2}e_{i,k} - r_{i,k}\mathrm{sig}^{\gamma}(e_{i,k})\\ &\quad + \tilde{\Phi}_{i,k}S_{i,k}) + \frac{1}{2}e_{i,k+1}^2 + \frac{1}{2}\chi_{i,k+1}^2 + \chi_{i,k}\dot{\chi}_{i,k}\\ &\quad - \tilde{\Phi}_{i,k}\dot{\hat{\Phi}}_{i,k} + \frac{1}{4}\tilde{\Phi}_{i,k}^2\end{aligned} \tag{35}$$

The adaptive law of $\dot{\hat{\Phi}}_{i,k}$ is designed as

$$\dot{\hat{\Phi}}_{i,k} = e_{i,k}S_{i,k} - \Gamma_{i,k}\hat{\Phi}_{i,k} \tag{36}$$

where $\Gamma_{i,k}$ is positive design constant.

Considering (17), (18), (36), one has

$$\begin{aligned}\dot{V}_{i,k} &\le e_{i,k}(-k_{i,k}e_{i,k} - \frac{1}{2}e_{i,k} - r_{i,k}\mathrm{sig}^{\gamma}(e_{i,k})) + \chi_{i,k}\dot{\chi}_{i,k}\\ &\quad + \frac{1}{2}e_{i,k+1}^2 + \frac{1}{2}\chi_{i,k+1}^2 + \Gamma_{i,k}\tilde{\Phi}_{i,k}^T\hat{\Phi}_{i,k} + \frac{1}{4}\tilde{\Phi}_{i,k}^2\\ &\quad + \Xi_{i,k}) + \Gamma_{i,k}\tilde{\Phi}_{i,k}^T\hat{\Phi}_{i,k}\\ &\le -(k_{i,k} + \frac{1}{2})e_{i,k}^2 - \frac{\chi_{i,k}^2}{\varpi_{i,k}} - r_{i,k}|e_{i,k}|^{\gamma+1}\\ &\quad - \frac{|\chi_{i,k}|^{\gamma+1}}{\varpi_{i,k}} + \frac{1}{2}e_{i,k+1}^2 + \frac{1}{2}\chi_{i,k+1}^2 + \frac{1}{4}\tilde{\Phi}_{i,k}^2\\ &\quad + \chi_{i,k}\Xi_{i,k} + \Gamma_{i,k}\tilde{\Phi}_{i,k}^T\hat{\Phi}_{i,k}\end{aligned} \tag{37}$$

where $\Xi_{i,k}(\cdot) = -\dot{\alpha}_{i,k-1}$ is continuous functions.

Step n_i: Define

$$V_{i,n_i} = \frac{1}{2}e_{i,n_i}^2 + \frac{1}{2}\chi_{i,n_i}^2 + \frac{1}{2}\tilde{\Phi}_{i,n_i}^2 \tag{38}$$

where $\tilde{\Phi}_{i,n_i} = \Phi_{i,n_i} - \hat{\Phi}_{i,n_i}$ are the estimation errors of NNs weight vector.

Then, we can obtain

$$\begin{aligned}\dot{V}_{i,n_i} &= e_{i,n_i}\dot{e}_{i,n_i} + \chi_{i,n_i}\dot{\chi}_{i,n_i} - \tilde{\Phi}_{i,n_i}\dot{\hat{\Phi}}_{i,n_i}\\ &= e_{i,n_i}(u_i + \hat{\Phi}_{i,n_i}^T S_{i,n_i} + \tilde{\Phi}_{i,n_i}S_{i,n_i} - \tilde{\Phi}_{i,n_i}S_{i,n_i}\\ &\quad + l_{i,n_i}\bar{e}_{i,1} - \dot{\pi}_{i,n_i}) + \chi_{i,n_i}\dot{\chi}_{i,n_i} - \tilde{\Phi}_{i,n_i}\dot{\hat{\Phi}}_{i,n_i}\end{aligned} \tag{39}$$

Next, based on the event-triggered control strategy, the event triggering mechanism is established as follows:

$$u_i(t) = \omega_i(t_k) \tag{40}$$

$$t_{k+1} = \inf\{t > t_k | \mathrm{abs}(\varsigma_i(t)) \ge \beta_i \mathrm{abs}(u_i(t)) + m_i\} \tag{41}$$

where $\omega_i(t)$ is the control law to be designed, $t_k > 0$ is the update time, $\varsigma_i(t) = \omega_i(t) - u_i(t)$ is the measurement error, $\beta_i \subset (0, 1)$ and $m_i > 0$.

Remark 2: During the time $t \in [t_k, t_{k+1})$, the control signal $u_i(t)$ holds as a constant $\omega_i(t_k)$, once the trigger model (41) is triggered, the control signal $u_i(t) = \omega_i(t_{k+1})$ will be obtained. According to (41), if we choose $|\lambda_{i1}(t)| \le 1$ and $|\lambda_{i2}(t)| \le 1$, we know that

$$\omega_i(t) = (1 + \lambda_{i1}(t)\beta_i)u_i(t) + \lambda_{i2}(t)m_i \tag{42}$$

that is

$$u_i(t) = \frac{\omega_i(t)}{1 + \lambda_{i1}(t)\beta_i} - \frac{\lambda_{i2}(t)m_i}{1 + \lambda_{i1}(t)\beta_i} \tag{43}$$

According to (39) and (43), it can be obtained that

$$\begin{aligned}\dot{V}_{i,n_i} &= e_{i,n_i}\bigg(\frac{\omega_i(t)}{1 + \lambda_{i1}(t)\beta_i} - \frac{\lambda_{i2}(t)m_i}{1 + \lambda_{i1}(t)\beta_i} + \hat{\Phi}_{i,n_i}^T S_{i,n_i}\\ &\quad + \tilde{\Phi}_{i,n_i}S_{i,n_i} - \tilde{\Phi}_{i,n_i}S_{i,n_i} + l_{i,n_i}\bar{e}_{i,1} - \dot{\pi}_{i,n_i}\bigg)\\ &\quad + \chi_{i,n_i}\dot{\chi}_{i,n_i} - \tilde{\Phi}_{i,n_i}\dot{\hat{\Phi}}_{i,n_i}\end{aligned} \tag{44}$$

The event-triggered controller is given as follows:

$$\omega_i(t) = -(1 + \beta_i)\bigg(\alpha_{i,n_i}\tanh\bigg(\frac{e_{i,n_i}\alpha_{i,n_i}}{\epsilon_i}\bigg) + \bar{m}_i\tanh\bigg(\frac{e_{i,n_i}\bar{m}_i}{\epsilon_i}\bigg)\bigg) \tag{45}$$

where $\bar{m}_i > \frac{m_i}{1-\beta_i}$, $\epsilon_i > 0$.

Using (45), one can obtain that

$$\begin{aligned}&e_{i,n_i}\frac{\omega_i(t) - \lambda_{i2}(t)m_i}{1 + \lambda_{i1}(t)\beta_i}\\ &\le \frac{e_{i,n_i}\omega_i(t)}{1 + \lambda_{i1}(t)\beta_i} + \bigg|\frac{e_{i,n_i}\lambda_{i2}(t)m_i}{1 + \lambda_{i1}(t)\beta_i}\bigg|\\ &\le -e_{i,n_i}\alpha_{i,n_i}\tanh\bigg(\frac{e_{i,n_i}\alpha_{i,n_i}}{\epsilon_i}\bigg)\\ &\quad - e_{i,n_i}\bar{m}_i\tanh\bigg(\frac{-e_{i,n_i}\bar{m}_i}{\epsilon_i}\bigg) + \bigg|\frac{e_{i,n_i}m_i}{1-\beta_i}\bigg|\\ &\le e_{i,n_i}\alpha_{i,n_i} - |e_{i,n_i}\bar{m}_i| + \bigg|\frac{e_{i,n_i}m_i}{1-\beta_i}\bigg| + 0.557\epsilon_i\\ &\le e_{i,n_i}\alpha_{i,n_i} + 0.557\epsilon_i\end{aligned} \tag{46}$$

From (46), we can find that

$$\begin{aligned}\dot{V}_{i,n_i} &= e_{i,n_i}\bigg(\alpha_{i,n_i} + \hat{\Phi}_{i,n_i}^T S_{i,n_i} + \tilde{\Phi}_{i,n_i}S_{i,n_i} - \tilde{\Phi}_{i,n_i}S_{i,n_i}\\ &\quad + l_{i,n_i}\bar{e}_{i,1} - \dot{\pi}_{i,n_i}\bigg) + \chi_{i,n_i}\dot{\chi}_{i,n_i}\\ &\quad - \tilde{\Phi}_{i,n_i}\dot{\hat{\Phi}}_{i,n_i} + 0.557\epsilon_i\end{aligned} \tag{47}$$

Using (17), (18), (21), one can obtain that

$$\begin{aligned}\dot{V}_{i,n_i} &= -k_{i,n_i}e_{i,n_i}^2 - \frac{3}{2}e_{i,n_i}^2 - r_{i,n_i}e_{i,n_i}\mathrm{sig}^{\gamma}(e_{i,n_i})\\ &\quad + e_{i,n_i}\bigg(\tilde{\Phi}_{i,n_i}S_{i,n_i} - \tilde{\Phi}_{i,n_i}S_{i,n_i}\bigg)\\ &\quad + \chi_{i,n_i}\bigg(\frac{\alpha_{i,n_i-1} - \pi_{i,n_i} + \mathrm{sig}^{\gamma}(-\chi_{i,n_i})}{\varpi_{i,n_i}}\bigg)\\ &\quad - \tilde{\Phi}_{i,n_i}\dot{\hat{\Phi}}_{i,n_i} + 0.557\epsilon_i\end{aligned} \tag{48}$$

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Consider

$$-e_{i,n_i}\tilde{\Phi}_{i,n_i}S_{i,n_i} \le e_{i,n_i}^2 + \frac{1}{4}\tilde{\Phi}_{i,n_i}^2 \tag{49}$$

The adaptive law of $\dot{\hat{\Phi}}_{i,n_i}$ is designed as

$$\dot{\hat{\Phi}}_{i,n_i} = e_{i,n_i}S_{i,n_i} - \Gamma_{i,n_i}\hat{\Phi}_{i,n_i} \tag{50}$$

where $\Gamma_{i,n_i} > 0$ are positive design constants.

Then, we have

$$\begin{aligned}\dot{V}_{i,n_i} \le& -(k_{i,n_i}+\frac{1}{2})e_{i,n_i}^2 - \frac{\chi_{i,n_i}^2}{\varpi_{i,n_i}} - r_{i,n_i}|e_{i,n_i}|^{\gamma+1}\\ &- \frac{|\chi_{i,n_i}|^{\gamma+1}}{\varpi_{i,n_i}} + \chi_{i,n_i}\Xi_{i,n_i} + \frac{1}{4}\tilde{\Phi}_{i,n_i}^2\\ &+ \Gamma_{i,n_i}\tilde{\Phi}_{i,n_i}^T\hat{\Phi}_{i,n_i}\end{aligned} \tag{51}$$

where $\Xi_{i,n_i}(\cdot) = -\dot{\alpha}_{i,n_i-1}$ is continuous functions.

Construct the Lyapunov function

$$V = \sum_{i=1}^{N}\sum_{q=1}^{n_i} V_{i,q} \tag{52}$$

Then, we have

$$\begin{aligned}\dot{V} \le \sum_{i=1}^{N}\Bigg\{ &-\sum_{q=1}^{n_i}k_{i,q}e_{i,q}^2 - \sum_{q=2}^{n_i}\left(\frac{1}{\varpi_{i,q}}-1\right)\chi_{i,q}^2\\ &+ \frac{1}{2}\xi_{i,1}^2 + \frac{1}{2}\bar{\xi}_{i,1}^2 + \sum_{q=2}^{n_i}\frac{1}{2}\sigma_{i,q}^2 - \sum_{q=1}^{n_i}r_{i,q}|e_{i,q}|^{\gamma+1}\\ &- \sum_{q=2}^{n_i}\frac{|\chi_{i,q}|^{\gamma+1}}{\varpi_{i,q}} + \sum_{q=1}^{n_i}\Gamma_{i,q}\tilde{\Phi}_{i,q}^T\hat{\Phi}_{i,q}\\ &+ \bar{\Gamma}_{i,1}\tilde{\Delta}_{i,1}^T\hat{\Delta}_{i,1}\Bigg\}\end{aligned} \tag{53}$$

where $|\Xi_{i,q}| \le \sigma_{i,q}$. From $\Gamma_{i,q}\tilde{\Phi}_{i,q}\hat{\Phi}_{i,q} \le -\frac{\Gamma_{i,q}(2o_i-1)}{2o_i}\tilde{\Phi}_{i,q}^2 + \frac{\Gamma_{i,q}o_i}{2}\Phi_{i,q}^2$, $\bar{\Gamma}_{i,1}\tilde{\Delta}_{i,1}\hat{\Delta}_{i,1} \le -\frac{\bar{\Gamma}_{i,1}(2p_i-1)}{2p_i}\tilde{\Delta}_{i,1}^2 + \frac{\bar{\Gamma}_{i,1}p_i}{2}\Delta_{i,1}^2$, $o_i > \frac{1}{2}$, $p_i > \frac{1}{2}$ and substituting it into (53), we have

$$\begin{aligned}\dot{V} \le \sum_{i=1}^{N}\Bigg\{ &-\sum_{q=1}^{n_i}k_{i,q}e_{i,q}^2 - \sum_{q=2}^{n_i}\left(\frac{1}{\varpi_{i,q}}-1\right)\chi_{i,q}^2\\ &+ \frac{1}{2}\xi_{i,1}^2 + \frac{1}{2}\bar{\xi}_{i,1}^2 + \sum_{q=2}^{n_i}\frac{1}{2}\sigma_{i,q}^2 - \sum_{q=1}^{n_i}r_{i,q}|e_{i,q}|^{\gamma+1}\\ &- \sum_{q=2}^{n_i}\frac{|\chi_{i,q}|^{\gamma+1}}{\varpi_{i,q}} - \sum_{q=1}^{n_i}\left(\frac{\Upsilon_{i,q}}{s_{i,q}}\tilde{\Phi}_{i,q}^2\right)^{\frac{\gamma+1}{2}}\\ &+ \sum_{q=1}^{n_i}\left(\frac{\Upsilon_{i,q}}{s_{i,q}}\tilde{\Phi}_{i,q}^2\right)^{\frac{\gamma+1}{2}} - \sum_{q=1}^{n_i}\frac{\Upsilon_{i,q}}{s_{i,q}}\tilde{\Phi}_{i,q}^2 + \sum_{q=1}^{n_i}\Gamma_{i,q}o_i\Phi_{i,q}^2\\ &- \left(\frac{\bar{\Upsilon}_{i,1}}{\psi_{i,1}}\tilde{\Delta}_{i,1}^2\right)^{\frac{\gamma+1}{2}} + \left(\frac{\bar{\Upsilon}_{i,1}}{\psi_{i,1}}\tilde{\Delta}_{i,1}^2\right)^{\frac{\gamma+1}{2}} - \frac{\bar{\Upsilon}_{i,1}}{\psi_{i,1}}\tilde{\Delta}_{i,1}^2\\ &+ \bar{\Gamma}_{i,1}p_i\Delta_{i,1}^2 + 0.557\epsilon_i\Bigg\}\end{aligned} \tag{54}$$

where $\Upsilon_{i,q} = s_{i,q}[(\Gamma_{i,q}(2o_i-1))/2o_i]$, $\bar{\Upsilon}_{i,1} = \psi_{i,1}[(\bar{\Gamma}_{i,1}(2p_i-1))/2p_i]$. Thus, we have

$$\dot{V} \le -pV - qV^{\frac{\gamma+1}{2}} + \nu \tag{55}$$

where $p = \min_{1\le i\le N}\left\{2k_{i,q}, 2\left(\frac{1}{\varpi_{i,q}}-1\right), 2\Upsilon_{i,q}, 2\bar{\Upsilon}_{i,1}\right\}$, $q = \min_{1\le i\le N}\left\{r_{i,q}2^{(1+\gamma)/2}, 2^{(1+\gamma)/2}\frac{1}{\varpi_{i,q}}, (2\Upsilon_{i,q})^{(1+\gamma)/2}, (2\bar{\Upsilon}_{i,1})^{(1+\gamma)/2}\right\}$, and

$$\begin{aligned}\nu = \sum_{i=1}^{N}\Bigg\{ &\frac{1}{2}\xi_{i,1}^2 + \frac{1}{2}\bar{\xi}_{i,1}^2 + \sum_{q=2}^{n_i}\frac{1}{2}\sigma_{i,q}^2 + \sum_{q=1}^{n_i}\Gamma_{i,q}o_i\Phi_{i,q}^2\\ &+ \bar{\Gamma}_{i,1}p_i\Delta_{i,1}^2 + 0.557\epsilon_i\Bigg\}\end{aligned} \tag{56}$$

Then, there exists a constant $0 < \beta < 1$ such that $\dot{V} \le -pV - qV^{\frac{\gamma+1}{2}} + \nu$ can be expressed as

$$\dot{V} \le -\beta pV - (1-\beta)pV - qV^{\frac{\gamma+1}{2}} + \nu \tag{57}$$

or

$$\dot{V} \le -pV - \beta qV^{\frac{\gamma+1}{2}} - (1-\beta)qV^{\frac{\gamma+1}{2}} + \nu \tag{58}$$

From (57), we have $\dot{V} \le -\beta pV - qV^{\frac{\gamma+1}{2}}$ if $V > [\nu/(1-\beta)p]$. Then, according to Lemma 1, we can obtain that V in finite time will drive $e_{i,1}, e_{i,q}, \chi_{i,q+1}, \tilde{\theta}_{i,q}$ into the region

$$(e_{i,q}, \chi_{i,q}, \tilde{\Phi}_{i,q}, \tilde{\Delta}_{i,1}) \in \left\{V \le \frac{\nu}{(1-\beta)p}\right\} \tag{59}$$

The time needed to arrive (57) is given as $T_r \le [1/(\beta p(1-[(1+\gamma)/2]))]\ln[(\beta pV^{1-[(1+\gamma)/2]}(0)+q)/q]$. From (58), we have $\dot{V} \le -pV - \beta qV^{\frac{\gamma+1}{2}}$ if $V^{\frac{\gamma+1}{2}} > [\nu/(1-\beta)q]$. Then, we also known that $e_{i,1}, e_{i,q+1}, \chi_{i,q+1}, \tilde{\theta}_{i,q}$ into the region

$$(e_{i,q}, \chi_{i,q}, \tilde{\Phi}_{i,q}, \tilde{\Delta}_{i,1}) \in \left\{V \le \left(\frac{\nu}{(1-\beta)b}\right)^{\frac{2}{\gamma+1}}\right\} \tag{60}$$

in finite time $T_r \le [1/(p(1-[(1+\gamma)/2]))]\ln[(pV^{1-[(1+\gamma)/2]}(0)+\beta q)/\beta q]$. Thus, from the above analysis, we know that $|e_{i,1}| \le \min\left\{\sqrt{[2\nu/(1-\beta)p]}, \sqrt{2[\nu/(1-\beta)q]^{[2/(\gamma+1)]}}\right\}$ in finite time $T_1 = \max\left\{1/(\beta p(1-[(1+\gamma)/2]))\ln[(\beta pV^{1-[(1+\gamma)/2]}(0)+q]/q, 1/(p(1-[(1+\gamma)/2]))\ln[(pV^{1-[(1+\gamma)/2]}(0)+\beta q]/\beta q\right\}$.

Next, we prove that the adaptive output-feedback neural bipartite control scheme with event-triggered can effectively exclude Zeno behavior [42]. To validate this, we can find a $t^0 > 0$ satisfying $\forall k \in z^+$, $\{t_{k+1}-t_k\} \ge t^0$. Knowing the $\varsigma_i(t) = \omega_i(t) - u_i(t)$, $\forall t \in [t_k, t_{k+1})$, we can obtain

$$\frac{d}{dt}|\varsigma_i| = \frac{d}{dt}(\varsigma_i\cdot\varsigma_i)^{\frac{1}{2}} = \text{sign}(\varsigma_i)\dot{\varsigma}_i \le |\dot{\omega}_i| \tag{61}$$

In conjunction with (45), ω_i is differentiable and $\dot{\omega}_i$ is a function of all the bounded closed-loop signals. Consequently, there exists a constant $\varpi_i > 0$ satisfying $|\dot{\omega}_i(t)| \le \varpi_i$. Based on $\varsigma_i(t_k) = 0$ and $\lim_{t\to t_{k+1}}\varsigma_i(t) = \beta_i\text{abs}(u_i(t)) + m_i$, we can obtain the lower bound of interexecution intervals t^0 satisfy $t^0 \ge m_i/\varpi_i > 0$, which denotes that the Zeno-behavior does not happen.

In addition, we prove that the Compact set Z of RBF NNs is an invariant set. Based on the above mentioned analysis, it can be deduced that all variables are bounded and

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converge to the compact set $\Omega := \{e_{i,q}, \chi_{i,q}, \tilde{\Phi}_{i,q}, \Delta_{i,1}$for $q = 1, \ldots, n_i \ |V \leq M \triangleq \min\{M_1, M_2\}\}$, where $M_1 = \frac{v}{(1-\beta)p}$, $M_2 = (\frac{v}{(1-\beta)b})^{(2/(\gamma+1))}$. Thus, we known that $e_{i,j} \leq D_i \triangleq \min\{\sqrt{[2v/(1-\beta)p]}, \sqrt{2[v/(1-\beta)q]^{[2/(\gamma+1)]}}\}$. On the other hand, based on $e_{i,1} = x_{i,1} - \text{sign}(a_{i,j})x_{j,1}$, a path $P_i = \min\{(o_i, o_j), (o_j, o_{j_1}), \ldots, (o_{j_l}, o_0)\}$, and by Assumption 2, we can obtain that $|x_{i,1}| \leq |r(t)|+|e_{i,1}|+|e_{j,1}|+\ldots+|e_{j_l,1}| \leq Y_0+\sum_{s\in\{l|o_l\in P_i\}} D_s = K_{i1}$. Then, $|\hat{x}_{i,2}| \leq |e_{i,2}|+|\chi_{i,2}|+|\alpha_{i,1}|$, we known from the definition of $\alpha_{i,1}$ in (19), $\alpha_{i,1}$ is a function of $x_{i,1}$, x_{j_1}, $\hat{\Phi}_{i,1}$, $\hat{\Delta}_{i,1}$. The boundedness of $x_{i,1}$, $x_{j,1}$, $\hat{\Phi}_{i,1}$, $\hat{\Delta}_{i,1}$ indicate that $\alpha_{i,1}$ is bounded and there exists a constant $\bar{L}_{i1}$ satisfies $|\alpha_{i,1}| \leq \bar{L}_{i1}$. Thus, it can be concluded that $|\hat{x}_{i,2}| \leq 2D_i + \bar{L}_{i1} = K_{i2}$. Likewise, it can also be proved that $|\hat{x}_{i,j}| \leq K_{ij}$. In this way, all states of the closed loop system will remain in the set $\Omega_{x_i} := \{x_i||x_{i,1}| \leq K_{i1}, |\hat{x}_{i,l}| \leq K_{il}, 2 \leq l \leq n_i\}$ if $x_i(t_0)$ stay in the set Ω_{x_i}. It should be pointed out that $x_i = [x_{i,1}, \hat{x}_{i,2}, \ldots, \hat{x}_{i,n_i}]$ are the inputs of NNs $\Phi_{i,l}^T S_{i,l}(\bar{x}_{i,l})$ in our paper, and Ω_{x_i} can be regarded as Ω_Z, which is an invariant set. In general, it is not easy to determine which sets of neural networks are active when they are used to estimate nonlinear functions. However, we found through reading a lot of references, e.g., [49, 53, 54], that the compact set of neural network exists and can be estimated, so we constructed the compact set represented by Ω_{x_i} for the neural network in this paper, and all the state inputs of NNs will remain in the set Ω_{x_i} as long as the initial state inputs remain in the set Ω_{x_i}. In fact, many important results also default to this and no explicit scope for these sets is given. ■

Remark 3: Similar to [50-52], $\Xi_{i,k}(\cdot)$, $\Xi_{i,n_i}(\cdot)$ in this paper are continuous functions with $\Xi_{i,2}(\cdot) = -\frac{\partial\alpha_{i,1}}{\partial x_{i,1}}\dot{x}_{i,1} - \frac{\partial\alpha_{i,1}}{\partial x_{j,1}}\dot{x}_{j,1} - \frac{\partial\alpha_{i,1}}{\partial\hat{\Phi}_{i,1}}\dot{\hat{\Phi}}_{i,1} - \frac{\partial\alpha_{i,1}}{\partial\hat{\Delta}_{i,1}}\dot{\hat{\Delta}}_{i,1}$, $\Xi_{i,j}(\cdot) = -\frac{\partial\alpha_{i,j}}{\partial x_{i,1}}\dot{x}_{i,1} - \sum_{k=2}^{j}\frac{\partial\alpha_{i,k}}{\partial\hat{x}_{i,k}}\dot{\hat{x}}_{i,k} - \frac{\partial\alpha_{i-1}}{\partial x_{j,1}}\dot{x}_{j,1} - \frac{\partial\alpha_{i,j}}{\partial\hat{\Phi}_{i,j}}\dot{\hat{\Phi}}_{i,j}$. Moreover, Â for any initial $V(0) \leq M$, the set $\Omega_1 := \{V(t) \leq M\} = \{e_{i,1}^2 + \tilde{\Phi}_{i,1}^2 + \tilde{\Delta}_{i,1}^2 + e_{i,2}^2 + \tilde{\Phi}_{i,2}^2 + \chi_{i,2}^2 + \ldots + e_{i,n_i}^2 + \tilde{\Phi}_{i,n_i}^2 + \chi_{i,n_i}^2 \leq 2M\}$ are the compact in R^{3n}. Thus, there exist positive constants $\sigma_{i,k}$ such that $|\Xi_{i,k}(\cdot)| \leq \sigma_{i,k}$ on Ω_1. It is necessary to point out that the values of $\sigma_{i,k}$ are unknown, the boundedness of $\Xi_{i,k}(\cdot)$ only need to satisfy during the stability analysis in this paper.

Remark 4: The event-triggered mechanism and bipartite consensus protocol are co-designed to reduce the update times of the controller, and the key ideal in this work is to integrate the prioritized strategy into the bipartite consensus tracking control. According to prioritized strategy, the main idea is to reconstruct the topological relationship between agent systems. Thus, the consensus protocol can be designed based on information classification and the communication time and interference between the leader and agents can be reduced.

Remark 5: In this paper, the strict limitation to a directed signed graph be structurally balanced is removed by using prioritized strategy. Besides, each agent only require obtaining its unique parent node's information by using prioritized strategy, which greatly alleviate the computing burden in our proposed method. In the future, some problems still need to be further investigated about this topic, e.g., how to design the fixed-time [2], [50] bipartite consensus protocol for nonlinear MASs under the idea of the proposed prioritized strategy, and the bipartite consensus control problem based on prioritized strategy for nonlinear MASs considering cyber attacks [51] will be investigated.

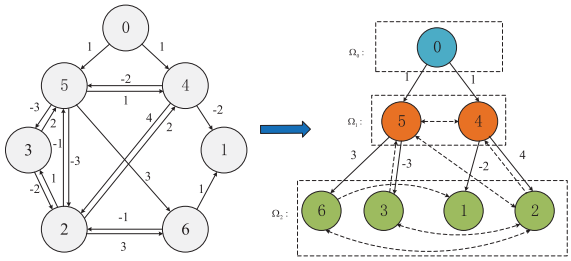


Fig. 2. Communications structurally unbalanced topology.

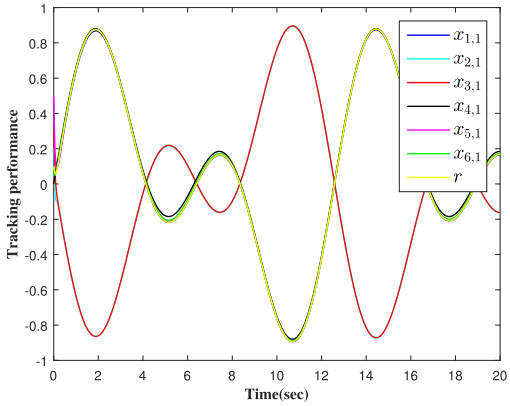


Fig. 3. Bipartite consensus performance.

IV. Simulation Studies

Example 1: Consider a network under structurally unbalanced directed signed graph in Fig. 2. The dynamics of each agents are given as:

$$\begin{cases} \dot{x}_{i,1} = x_{i,2} + f_{i,1}(x_{i,1}) \\ \dot{x}_{i,2} = u_i + f_{i,2}(\bar{x}_i) \\ y_i = x_{i,1} \end{cases} \tag{62}$$

The nonlinear functions are chosen as $f_{1,1} = 0.1x_{1,1}$, $f_{1,2} = -0.1x_{1,1}x_{1,2}$. $f_{2,1} = -0.1x_{2,1}$, $f_{2,2} = 0.1x_{2,2}$. $f_{3,1} = 0.1x_{3,1}$, $f_{3,2} = -0.1x_{3,1}x_{3,2}$. $f_{4,1} = 0.1x_{4,1}$, $f_{4,2} = 0.1x_{4,2}$. $f_{5,1} = 0.1x_{5,1}$, $f_{5,2} = 0.1x_{5,2}$. $f_{6,1} = 0.1x_{6,1}$, $f_{6,2} = 0.1x_{6,2}$, and the output of leader is chosen as $r(t) = 0.5(\sin t + \sin(0.5t))$.

Here, the conventional gauge transformation cannot employed to this topology. However, the nodes in Fig. 2 can achieve a bipartition based prioritized strategy in this paper. Fig. 2 is given to show the evolution of connected structurally unbalanced topology by using prioritized strategy. From Fig. 2, the shortest paths for agents $o_1 - o_6$ are $P_1 = \{(o_1, o_4), (o_4, o_0)\}$, $P_2 = \{(o_2, o_4), (o_4, o_0)\}$, $P_3 = \{(o_3, o_5), (o_5, o_0)\}$, $P_4 = \{(o_4, o_0)\}$, $P_5 = \{(o_5, o_0)\}$ and $P_6 = \{(o_6, o_5), (o_5, o_0)\}$, respectively. Besides, $S_{P_1} = -1$, $S_{P_2} = 1$, $S_{P_3} = -1$, $S_{P_4} = 1$, $S_{P_5} = 1$, $S_{P_6} = 1$. Then, two different sets can be obtained by using prioritized strategy, i.e., $O_1 = \{o_0, o_2, o_4, o_5, o_6\}$ and $O_2 = \{o_1, o_3\}$.

In simulation, the initial states of the six follower agents are selected as $x_1(0) = [0.1, 0]^T$, $x_2(0) = [0.2, 0.1]^T$, $x_3(0) = [0.3, 0.2]^T$, $x_4(0) = [0.5, 0.2]^T$, $x_5(0) = [0.1, 0.2]^T$, $x_6(0) = [0.1, 0.15]^T$ for all agents. The design parameters are chosen as $k_{i,1} = 10$, $k_{i,2} = 1$. $r_{i,q} = 2$, $s_{i,q} = 2$, $\tau_{i,q} = 0.01$,

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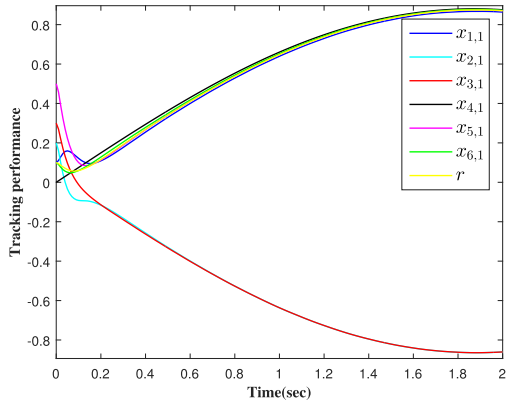


Fig. 4. Enlarged view of Bipartite consensus for $0 \leq t \leq 2$.

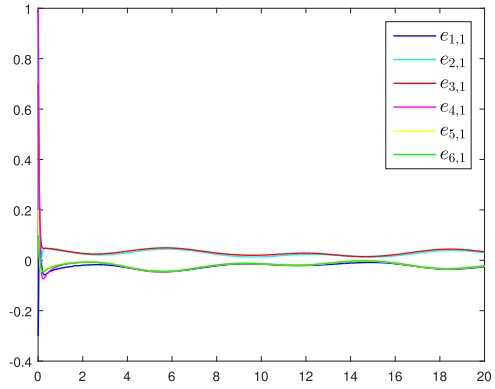


Fig. 5. Distributed bipartite errors.

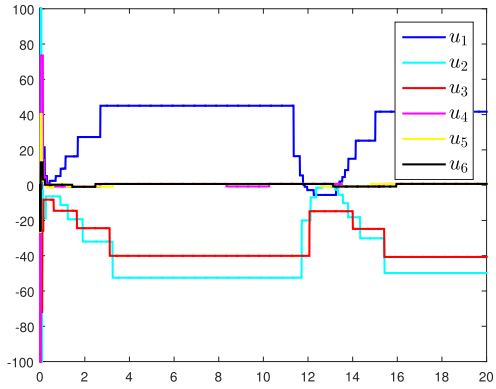


Fig. 6. The control signals.

$\Gamma_{i,q} = 1$, $\gamma = 0.8$, $\beta_i = 0.5$, $\bar{m}_i = 3$ for $i = 1, \ldots, 6$ and $q = 1, 2$. Simulation results are shown in Figs. 3-8, respectively.

Example 2: Consider the leader-following bipartite consensus tracking for a group of forced damped pendulums (FDPs) which can be described by [52]:

$$\ddot{\vartheta}_i = \frac{b_i}{m_i l_i^2}\dot{\vartheta}_i - \frac{c_i g}{l_i}\sin(\vartheta_i) + u_i, \quad i = 1, 2, 3 \tag{63}$$

where $g = 9.8$, $l_i = 1$, $b_i = -0.25$, $c_i = 9.8^{-1}$, $p_{i,1} = p_{i,2} = 1$, $\vartheta_1(0) = 0.2$, $\dot{\vartheta}_1(0) = 0.2$, $\vartheta_2(0) = 0.3$, $\dot{\vartheta}_2(0) = 0.3$, $\vartheta_3(0) = 0.4$, $\dot{\vartheta}_3(0) = 0.4$. The values and physical meaning of parameters can be found in [52]. Let $x_{i,1} = \vartheta_i$, $x_{i,2} = \dot{\vartheta}_i$,

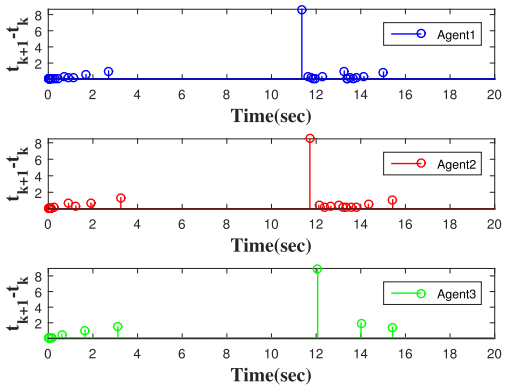


Fig. 7. The triggering instants of the each agent.

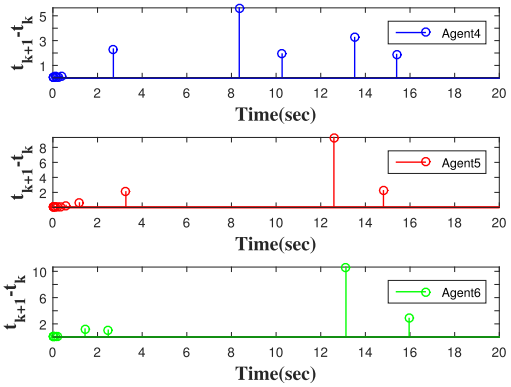


Fig. 8. The triggering instants of the each agent.

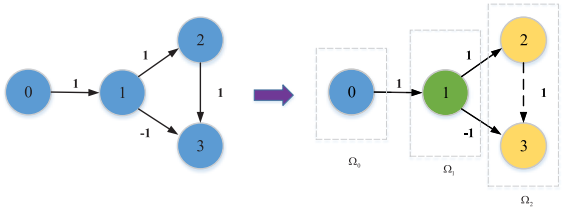


Fig. 9. Communications structurally unbalanced topology.

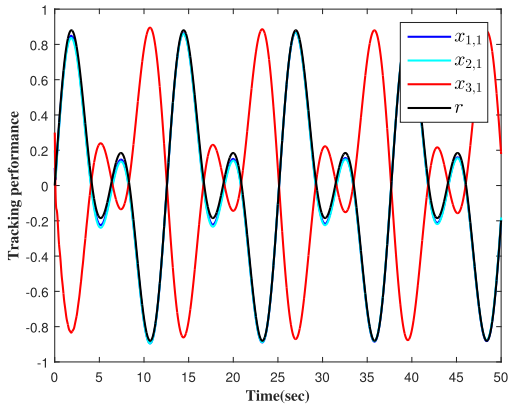


Fig. 10. Bipartite consensus performance.

then the system model can be rewritten as:

$$\begin{aligned} \dot{x}_{i,1} &= x_{i,2} \\ \dot{x}_{i,2} &= -\frac{b_i}{m_i l_i^2}x_{i,2} - \frac{c_i g}{l_i}\sin(x_{i,1}) + u_i \\ y_i &= x_{i,1}, i = 1, 2, 3 \end{aligned} \tag{64}$$

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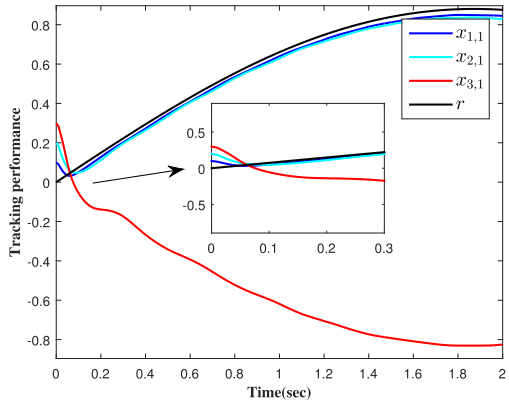


Fig. 11. Enlarged view of Bipartite consensus for $0 \le t \le 2$.

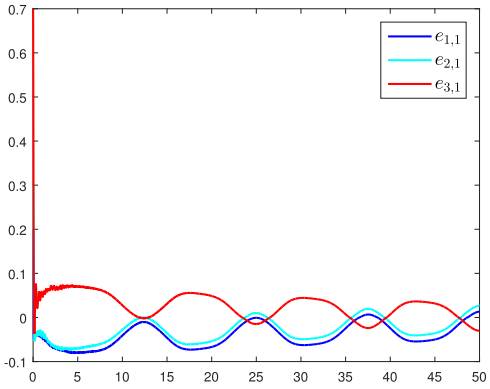


Fig. 12. Distributed bipartite errors.

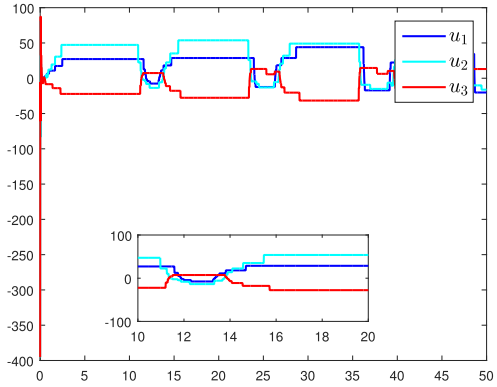


Fig. 13. The control signals.

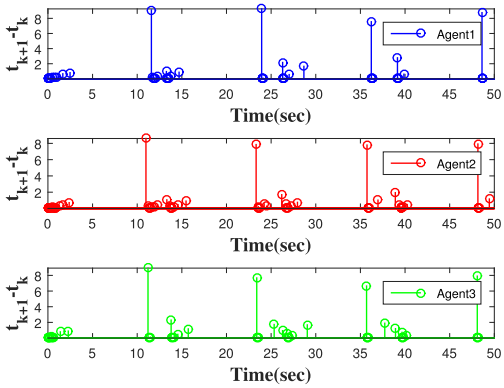


Fig. 14. The triggering instants of the each agent.

The communication graphs are considered in Fig. 9, and the output of leader is chosen as $r(t) = 0.5(\sin t + \sin(0.5t))$. Fig. 9 is given to show the evolution of connected structurally unbalanced topology by using prioritized strategy. From Fig. 9, the shortest pathes for agents $o_1 - o_3$ are $P_1 = \{(o_1, o_0)\}$, $P_2 = \{(o_2, o_1), (o_1, o_0)\}$ and $P_3 = \{(o_3, o_1), (o_1, o_0)\}$, respectively. Besides, $S_{P_1} = 1$, $S_{P_2} = 1$, $S_{P_3} = -1$. Then, two different sets can be obtained by using prioritized strategy, i.e., $O_1 = \{o_0, o_1, o_2\}$ and $O_2 = \{o_3\}$.

In simulation, the initial states of the three follower agents are selected as $x_1(0) = [0.1, 0]^T$, $x_2(0) = [0.2, 0.1]^T$, $x_3(0) = [0.3, 0.2]^T$ for all agents. The design parameters are chosen as $k_{i,1} = 10$, $k_{i,2} = 1$. $r_{i,q} = 2$, $s_{i,q} = 2$, $\tau_{i,q} = 0.01$, $\Gamma_{i,q} = 1$, $\gamma = 0.8$, $\beta_i = 0.5$, $\bar{m}_i = 3$ for $i = 1, \ldots, 3$ and $q = 1, 2$.

Simulation results are shown in Figs. 10-14, respectively. Figs. 10 - 11 show output trajectories of all agents. From Fig. 11, it is easy to obtain that bipartite consensus performance between followers and leader is fairly good obtained in a finite-time. As expressed in Fig. 12, the distributed bipartite consensus errors rapidly converge to compact a set in less than one second. Fig. 13 displays the trajectories of $u_i (i = 1, \ldots, 3)$ under the proposed protocol. Fig. 14 shows the time intervals of events.

V. Conclusion

In this paper, the neural state estimator-based event-triggered finite-time bipartite tracking control problem of nonlinear MASs has been solved. Based on prioritized strategy, the strict limitation to a directed signed graph be structurally balanced has been removed. The event-triggered mechanism and distributed dynamic surface control law have been co-designed by using the NNs and the backstepping method. Moreover, it has been guaranteed that all the signals are bounded, and the distributed consensus tracking errors can converge to a small neighborhood of the origin at a finite-time. Finally, the simulation results illustrated the validity of the proposed schemes.

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